

DocuMind Handbook: Retrieval-Augmented Generation

DocuMind is a small reference system for asking questions about documents.

Retrieval-augmented generation, often called RAG, retrieves relevant text before asking a language model to write an answer. This handbook is a self-contained sample document for reproducible evaluation experiments.

Ingestion and chunking

The ingestion pipeline extracts PDF text page by page and treats a text or Markdown file as one logical page. Text is split into overlapping character chunks so that a fact near a boundary remains discoverable. The default chunk size is 800 characters and the default overlap is 100.

Embeddings and storage

DocuMind embeds every chunk with all-MiniLM-L6-v2 from sentence-transformers.

The vectors and their metadata are stored in a persistent local ChromaDB collection. Metadata records the source filename, one-indexed page number, and chunk index so that answers can display useful citations.

Retrieval modes

Vector retrieval compares the question embedding with stored chunk vectors.

Hybrid retrieval combines vector results with lexical BM25 results.

Reciprocal Rank Fusion, or RRF, gives each result $1 \text{ divided by } k \text{ plus rank}$ and rewards chunks that appear near the top of both ranked lists.

Grounded answers

The answer prompt instructs the model to use only the retrieved context.

A source citation uses the format [filename, p.N]. When the context does not contain an answer, the required response is: I couldn't find this in the documents. Showing the source chunk makes each citation inspectable.

Evaluation and operations

The evaluation harness measures retrieval hit rate at k and mean reciprocal rank, abbreviated MRR. It can also ask a configured language model to judge answer quality on a one-to-five scale. Results are saved as CSV for README tables. The local index can be cleared from the application sidebar.